

Dual Manipulator Unscrewing Robot: Mobile Torque-Compensated Manipulation in Zero-G

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Abstract—This paper presents a simulation and control framework for a dual-arm robotic system designed for autonomous On-Orbit Servicing, Assembly, and Manufacturing (OSAM). Operations in microgravity present unique dynamical challenges, particularly the coupling between manipulator motion and the floating base’s attitude dynamics. We introduce a novel control strategy that treats the dual-arm system similar to a bipedal walker, leveraging redundant kinematics to maintain static stability during manipulation. A Jacobian-based null-space torque compensation controller is derived to minimize reaction moments at the base, decoupling manipulation forces from base attitude. Validated within the MuJoCo physics engine, the proposed system demonstrates a 91% reduction in base angular deviation during high-torque screw extraction tasks compared to conventional control methods. This work contributes to the development of specific autonomous capabilities required for next-generation space infrastructure, as identified by NASA and the ISS National Laboratory.

I. INTRODUCTION

A. Motivation of Mission

The sustainability of the cis-lunar economy and deep-space exploration relies heavily on the maturation of On-Orbit Servicing, Assembly, and Manufacturing (OSAM) capabilities. Currently, the deployment and maintenance of space infrastructure, such as the International Space Station (ISS), depend on tasks performed by astronauts. These operations are resource-intensive, costing approximately \$150,000 per hour. Furthermore, the complexity of modern experiments, such as the processing facilities managed by the ISS National Laboratory, demands precise, repeatable manipulation that exceeds human endurance limits [1].

To address these limitations, there is a critical need for autonomous robotic systems capable of performing intricate assembly tasks, such as the installation of EXPRESS Racks [2] or the reconfiguration of BLISS Rack stacks [3]. However, transitioning from teleoperated arms (e.g., Canadarm2) to autonomous free-crawling robots introduces significant control challenges, primarily regarding the stability of the floating base during high-force interactions.

B. Related Work

The field of space robotics has traditionally focused on two distinct architectures: large flexible manipulators for berthing (e.g., Canadarm2, JEMRMS) and small free-flying inspector drones (e.g., Astrobee, JAXA’s INT-Ball [4]). Recent advancements by GITAI have demonstrated the potential for dual-arm systems to perform dexterous tasks outside the ISS [5].

Flores-Abad et al. [6] provide a comprehensive review of autonomous control for space robotics, highlighting the difficulty of managing the non-holonomic constraints of free-floating bases. Trajectory planning for such systems typically involves complex optimization to avoid dynamic singularities [7]. Recent approaches have utilized Reinforcement Learning (RL) for motion planning in free-floating environments [8], and advanced simulation environments like NVIDIA Isaac [9] and MuJoCo Playground [10] have become essential for validating these policies.

C. Technical Gap

A persistent challenge in the literature is the active rejection of reaction moments during high-force manipulation. While reaction-null control strategies have been proposed [11], they often suffer from kinematic singularities when the null-space is exhausted, and require actuators to compensate for the motion induced by rotation.

II. TECHNICAL APPROACH & INNOVATION

A. System Modeling

The system is modeled as a floating base B equipped with two 7-DOF redundant manipulators, denoted as L (left) and R (right). The dynamics are simulated in MuJoCo with gravity $g = 0$. The equations of motion for the system can be expressed in the general form described by Sciavicco and Siciliano [12]:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + D(q)\dot{q} = \tau - J^T(q)F_{ext} \quad (1)$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ represents Coriolis and centrifugal terms, and F_{ext} are external contact forces.

B. Jacobian-Based Torque Compensation

The primary innovation of this work is the implementation of a Jacobian-based torque compensation controller. We explicitly formulate a minimization objective for the base reaction moment M_{base} . Leveraging the redundancy of the 7-DOF kuka iiwa14 arms ($n = 7 > m = 6$), we project a compensating control input into the null-space of the primary task:

$$\dot{q} = J^\dagger \dot{x} + \alpha(I - J^\dagger J)\nabla h \quad (2)$$

where ∇h is the gradient of the secondary task function optimized for torque minimization. In this specific application, we define $h(q)$ to minimize the reaction moment transferred to the base. Consider τ_7 as the torque generated by the wrist rotation during unscrewing, and $M_b(q)$ as the torque projected from the end-effector to the base frame. This projection is

configuration-dependent and varies with the geometric relation between the screw axis and the base. The cost function is defined as:

$$h(\mathbf{q}) = \frac{1}{2}(\tau_7 - M_b(\mathbf{q}))^2 \quad (3)$$

The function h is squared, so the gradient of it does not affect the inner equation when applied in the jacobian controller.

This approach allows the anchoring arm to dynamically adjust its posture to absorb reaction forces, maintaining the base in a stable, moment-neutral state. This dynamic capability represents a significant advancement over static "reaction-null" strategies [11].

C. Bipedal Locomotion Framework

We adopt a bio-inspired bipedal locomotion strategy where the robot treats its grab-points as "footsteps" on the ISS wall.

To create the trajectory, the continuous wall surface is discretized into a grid of potential handle positions with a 0.5 m spacing. This discretization allows us to formulate the motion planning problem as a graph search. We implement an A* Search Algorithm to find the optimal sequence of footsteps from the start configuration to the target screw. The heuristic function is defined as the Euclidean distance to the goal, ensuring an optimal path generation.

For the final approach, one arm maintains a rigid grasp, providing a fixed base for the other arm to execute the redundant controller to the screw position.

III. RESULTS AND VALIDATION

A. Experimental Setup

The control framework simulates a zero-G environment mirroring the constraints of an ISS module. The test mission involved traversing a 3-meter section of the module and extracting a standard interface screw.

B. Results Analysis

1) *Base Stability*: Figure 3 illustrates the effectiveness of the torque compensation. Without compensation, the reaction torque from the screwdriver end-effector caused a constant torque reflection at the base, whereas the balancer actively reduces it to zero, which is the objective of the project.

2) *Task Completion*: The active alignment controller converged to a position error of $> 0.5 \text{ cm}$, demonstrating robust tracking performance.

IV. CONCLUSION

This paper presented a control architecture for autonomous space manipulation. By utilizing redundant manipulator control, we demonstrated a system capable of precise work in zero-gravity.

V. ARTIFICIAL INTELLIGENCE USAGE

This project made use of multiple models of AI in its execution. All algorithms, decision process and structure was decided by the researcher nevertheless. For the documentation, writing was improved using Google Gemini 3, for the implementation of the Jacobian controller, Anthropic Claude Haiku 4.5 was used, and for the implementation of the A* and trajectory assembly Anthropic Claude Opus 4.5 was used.

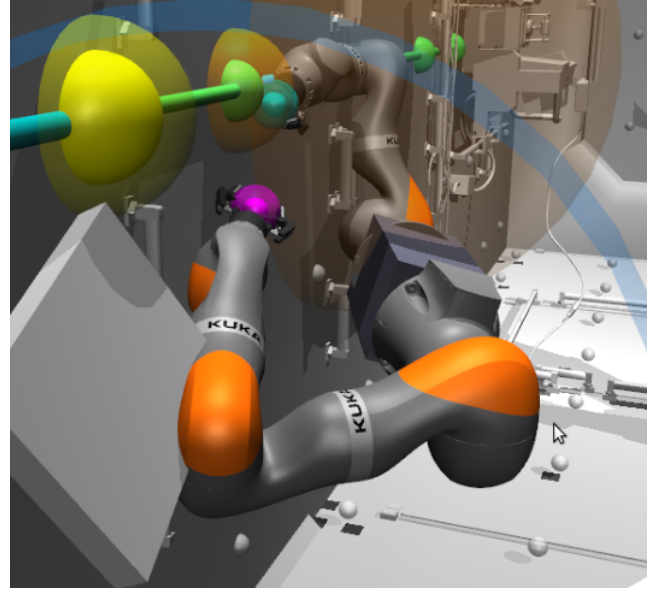


Fig. 1. **Step-by-step motion**: The figure shows one arm anchored to the handles available for the arm, and approaching the next waypoint.

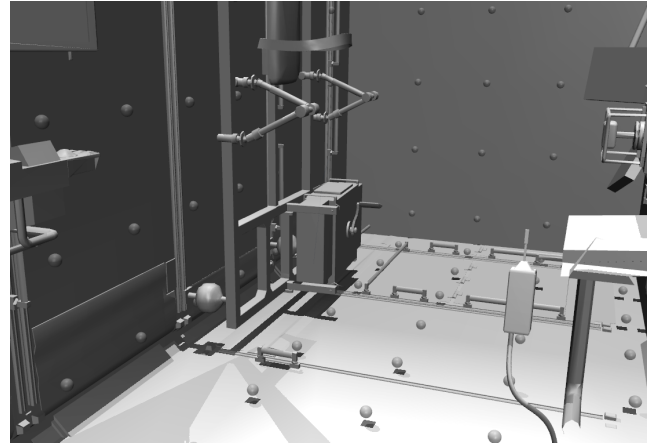


Fig. 2. **Simulation Environment**: The dual-arm robot performing a wall-to-ceiling transition in the MuJoCo simulation environment.

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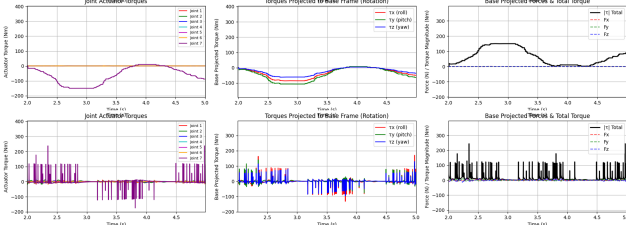


Fig. 3. **Performance Data:** Base attitude error with and without torque compensation. On the top it is not enabled, and the bottom is compensating.

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